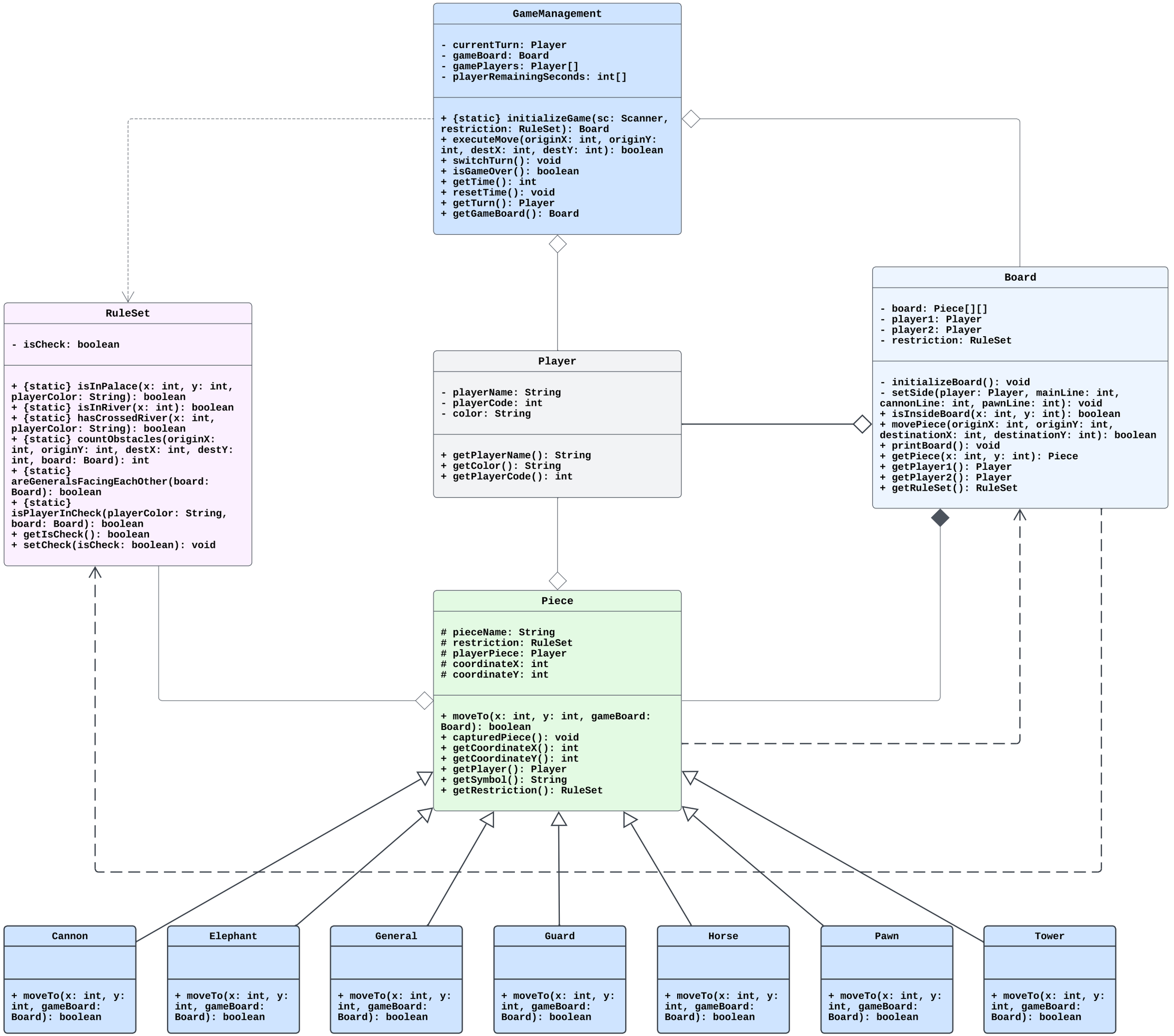
XIANQI


GameManagement
- currentTurn: Player
- gameBoard: Board
- gamePlayers: Player[]
- playerRemainingSeconds: int[]
+ {static} initializeGame(sc: Scanner, restriction: RuleSet): Board
+ executeMove(originX: int, originY: int, destX: int, destY: int): boolean
+ switchTurn(): void
+ isGameOver(): boolean
+ getTime(): int
+ resetTime(): void
+ getTurn(): Player
+ getGameBoard(): Board
RuleSet
- isCheck: boolean
+ {static} isInPalace(x: int, y: int, playerColor: String): boolean
+ {static} isInRiver(x: int): boolean
+ {static} hasCrossedRiver(x: int, playerColor: String): boolean
+ {static} countObstacles(originX: int, originY: int, destX: int, destY: int, board: Board): int
+ {static} areGeneralsFacingEachOther(board: Board): boolean
+ {static} isPlayerInCheck(playerColor: String, board: Board): boolean
+ getIsCheck(): boolean
+ setCheck(isCheck: boolean): void
Player
- playerName: String
- playerCode: int
- color: String
+ getPlayerName(): String
+ getColor(): String
+ getPlayerCode(): int
Board
- board: Piece[][]
- player1: Player
- player2: Player
- restriction: RuleSet
- initializeBoard(): void
- setSide(player: Player, mainLine: int, cannonLine: int, pawnLine: int): void
+ isInsideBoard(x: int, y: int): boolean
+ movePiece(originX: int, originY: int, destinationX: int, destinationY: int): boolean
+ printBoard(): void
+ getPiece(x: int, y: int): Piece
+ getPlayer1(): Player
+ getPlayer2(): Player
+ getRuleSet(): RuleSet
Piece
pieceName: String
restriction: RuleSet
playerPiece: Player
coordinateX: int
coordinateY: int
+ moveTo(x: int, y: int, gameBoard: Board): boolean
+ capturedPiece(): void
+ getCoordinateX(): int
+ getCoordinateY(): int
+ getPlayer(): Player
+ getSymbol(): String
+ getRestriction(): RuleSet
Cannon
+ moveTo(x: int, y: int, gameBoard: Board): boolean
Elephant
+ moveTo(x: int, y: int, gameBoard: Board): boolean
General
+ moveTo(x: int, y: int, gameBoard: Board): boolean
Guard
+ moveTo(x: int, y: int, gameBoard: Board): boolean
Horse
+ moveTo(x: int, y: int, gameBoard: Board): boolean
Pawn
+ moveTo(x: int, y: int, gameBoard: Board): boolean
Tower
+ moveTo(x: int, y: int, gameBoard: Board): boolean